

EE-482: Network Security and Cryptography

LECTURES NO 9: *Asymmetric Cryptographic Algorithms*

Engr. Tehseen Ahsan

**Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan**

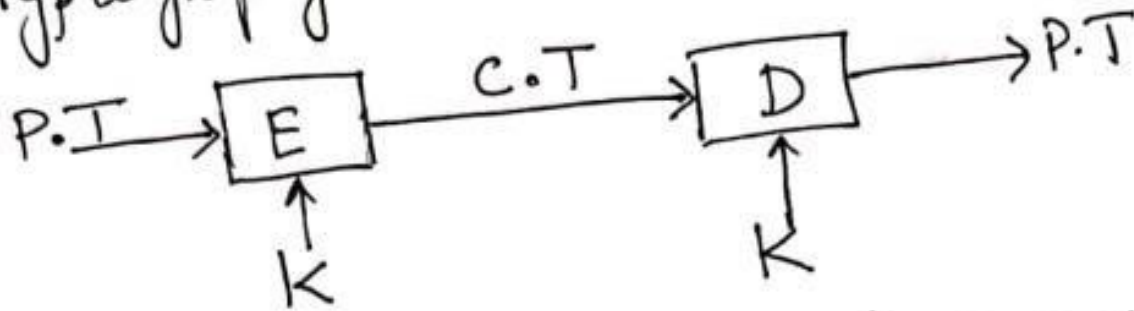
Organization of Lecture

1. Symmetric key model
2. Asymmetric key model
3. Characteristics of public key algorithm
4. Process (Encryption)
5. Digital signatures
6. Requirements for public key cryptography
7. Mathematical basis

1. Symmetric key model (1/2)

Symmetric Key Model

↳ It is also called as private key / secret key / conventional cryptography



- Algorithm requires same key for encryption & decryption.
- Encryption & Decryption are essentially identical (symmetric algorithm).

1. Symmetric key model (2/2)

Problems with Symmetric key schemes

- requires secure transmission of secret key
- In a network environment each pair of users has to have a different key $\Rightarrow$ too many keys.

$$\frac{n \cdot (n-1)}{2} \quad \text{key pairs}$$

Example 1: How many keys are required to implement symmetric algorithm with "12" participants?

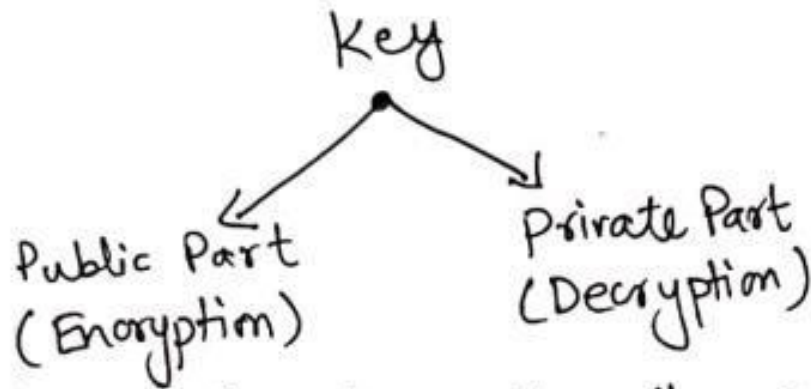
Ans: total no. of participants = $n = 12$

$$\text{total no. of keys with 12 participants} = \frac{n(n-1)}{2} = \frac{12(12-1)}{2} = 66$$

2. Asymmetric key model (1/4)

New Idea

Asymmetric key Cryptography } proposed by
or } Diffie-Hellman
Public key cryptography



- Based on mathematics rather than substitution/transposition schemes
- Different keys for encryption & decryption.
- One key is kept private (secret) other can be made public (not secret).
- Private and public keys are related mathematically.

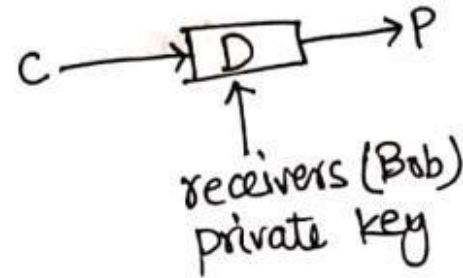
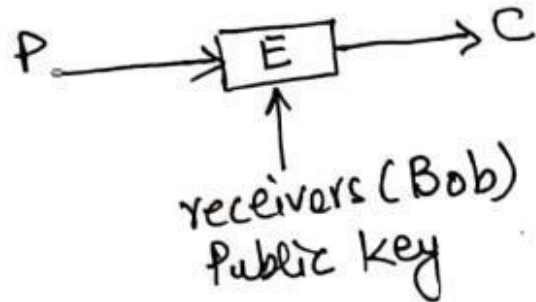
2. Asymmetric key model (2/4)

* Some Misconceptions

- 1) Public Key encryption is more secure than private-key encryption schemes
 - Security of any encryption scheme depends on
 - (a) length of key
 - (b) computational work in breaking a cipher.
 - public key is NOT better than private key
- 2) Public key encryption has made conventional schemes obsolete
 - DES is still widely used in various applications.
 - Symmetric key is much faster than public key.
 - Public key is used for authentication, digital signatures.
 - Symmetric key is used for bulk encryption e.g. high volume packet flow on network.

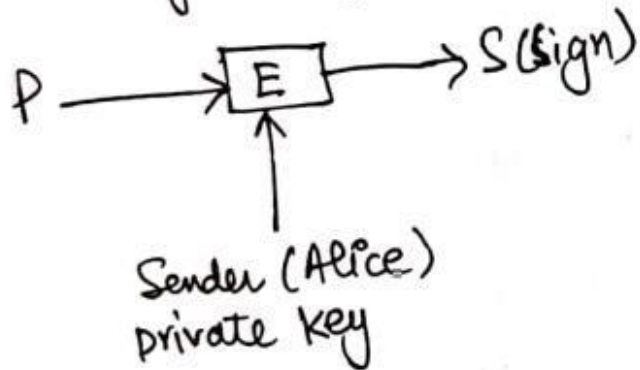
2. Asymmetric key model (3/4)

Confidentiality (Secrecy)

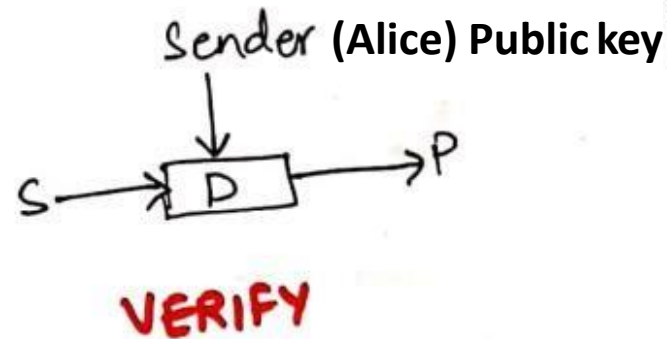


Authentication

→ Digital Signature

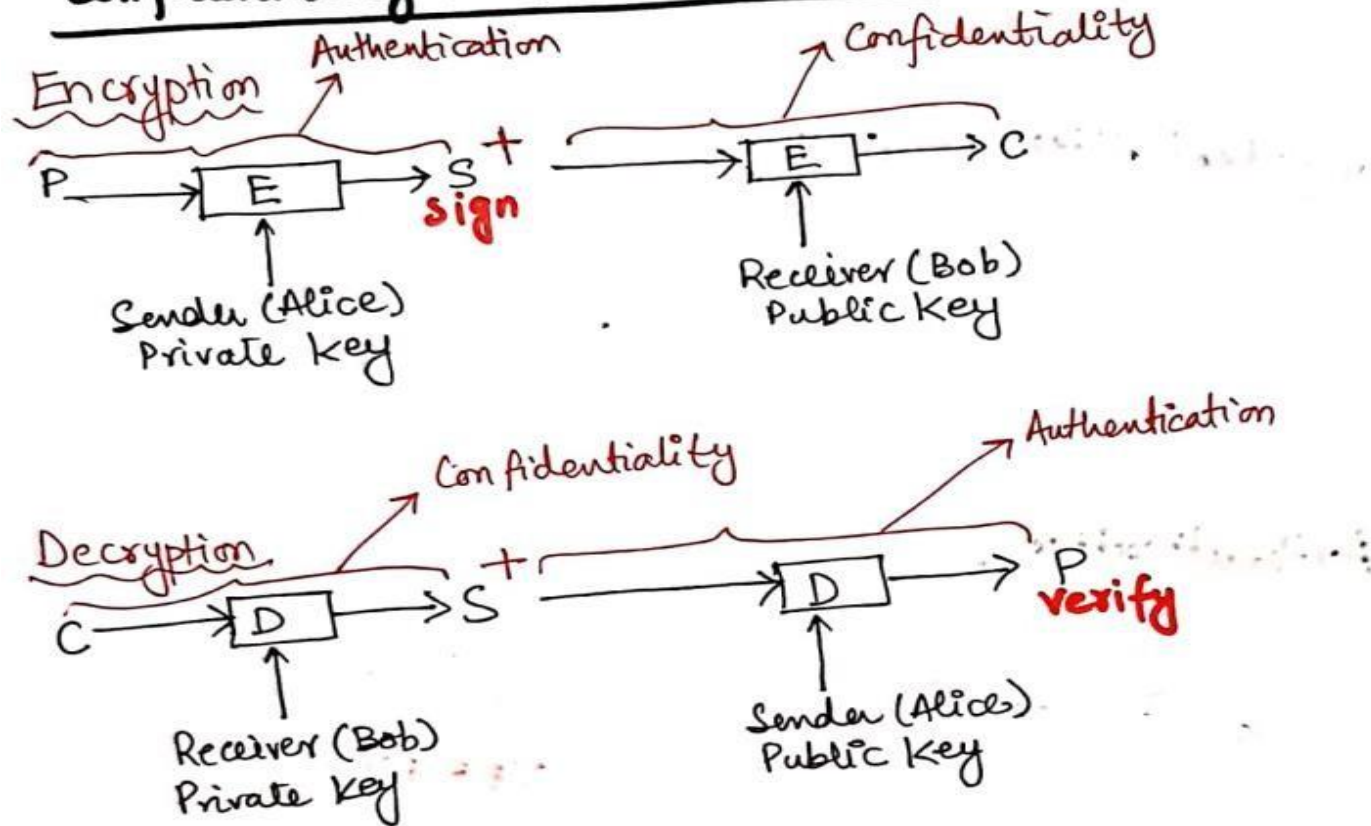


SIGN



2. Asymmetric key model (4/4)

Confidentiality with Authentication



Key Pair:

- Receiver (Public, Private)
- Sender (Public, Private)

3. Characteristics of public key algorithm

Characteristics of Public-Key Algorithm

- 1) It is computationally infeasible to determine decryption key (private key) given the knowledge of encryption key (public key) and the encryption algorithm.
- 2) Secrecy of asymmetric key method lies in secrecy of the private key.
- 3) It is computationally infeasible for an attacker, knowing the public key and C.T to recover the original P.T.
- 4) Either of the two related keys (key pair) may be used for encryption & decryption.

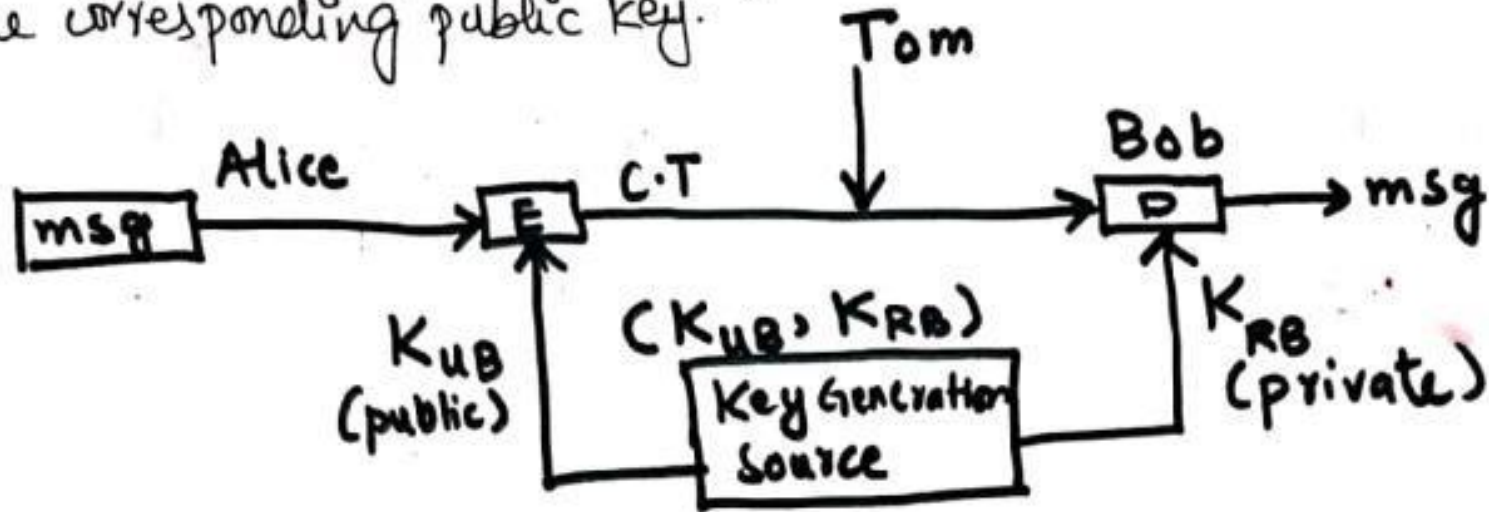
4. Process (Encryption) (1/2)

Process (Encryption)

- 1) Each end system in network generates a key pair (K_u, K_R) ; where K_u = public key & K_R = Private key.
- 2) Each system publishes its encryption (K_u) in a public register/file. The other key is kept a secret.
- 3) Alice wants to send Bob a message (let's assume)
- 4) Alice will pick Bob's public key from the register & encrypt the message using K_{uB} & sends the C.T.
- 5) Bob receives the C.T & decrypts it using his private key, K_{RB}
- 6) If Tom (attacker) intercepts the transmitted C.T; he doesn't have B's private key & cannot decrypt the C.T.

4. Process (Encryption) (2/2)

7. At any time the system can change its private key & publish the corresponding public key.



5. Digital Signatures

* Digital Signature

- Sender "signs" a message with its private key.
- Signing is achieved by a cryptographic algorithm applied to the message or to a small block of data
- Receiver "verifies" a message with the sender's public key

6. Requirements for public key cryptography

Requirements for Public Key Cryptography

- 1) Easy & efficient to generate keys (public & private).
- 2) Easy & efficient to encrypt and decrypt
- 3) Hard to compute private key from public key
- 4) Hard to compute plain text from public key & C.T.

7. Mathematical Basis (1/2)

Mathematical Basis

If we have a number that has only two factors (which are prime numbers), can generate a pair of keys.

Example: 10 has only two factors
2; 5 both prime.

if encrypt using 5 (as public)
only can decrypt using 2 (as private)

Nothing else (not even 5) can do decryption.

→ 10 is a small no. & with a small effort the attacker can break the scheme.

→ If no. is very large; difficult to break the scheme

7. Mathematical Basis (2/2)

One way function (f)

$$C = f(P) ; \text{ Easy}$$

$$P = f^{-1}(C) ; \text{ Infeasible}$$

Trap Door One way function (f)

$$C = f(K, P) ; \text{ easy if } K \& P \text{ known}$$

$$P = f^{-1}(K, C) ; \text{ infeasible if } K \text{ unknown, } C \text{ known}$$

$f = \text{function}$
 $f^{-1} = \text{Inverse function}$
 $C = C.T$
 $P = P.T$
 $K = \text{key}$

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*